



Numerical Investigation of Magnetohydrodynamic Jeffrey Fluid Flow with Nonlinear Thermal Radiation in a Vertical Channel

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Received: 09-29-2025

Revised: 12-12-2025

Accepted: 12-26-2025

Citation: A. B. Disu, W. Michael, C. Y. Ishola, and S. O. Ajibola, "Numerical investigation of magnetohydrodynamic Jeffrey fluid flow with nonlinear thermal radiation in a vertical channel," *Power Eng. Eng. Thermophys.*, vol. 4, no. 4, pp. 241–252, 2025. <https://doi.org/10.56578/peet040404>.



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Abstract: The magnetohydrodynamic flow of a Jeffrey fluid in a vertical channel was numerically investigated, with particular emphasis on the coupled effects of nonlinear thermal radiation, internal heat generation, variable viscosity, chemical reaction, and buoyancy forces. The governing partial differential equations describing momentum, heat, and mass transport were transformed into a system of ordinary differential equations through appropriate similarity transformations and the introduction of a stream function formulation. The resulting boundary value problem was solved numerically using the MATLAB bvp4c solver. The results show that the fluid velocity was significantly reduced with increasing magnetic field strength due to the enhancement of Lorentz forces, while the temperature distribution was elevated as a consequence of Joule heating. Nonlinear thermal radiation was found to substantially intensify the thermal field. Furthermore, an increase in the viscosity variation parameter and Jeffrey fluid parameters led to a suppression of both momentum and thermal boundary layers. In contrast, buoyancy forces were shown to accelerate the flow and enhance heat transfer. The influence of mass transfer was also critically examined, where an increase in the Schmidt number (Sc) and chemical reaction parameter resulted in a marked reduction in species concentration, thereby weakening mass diffusion. The interplay among these thermophysical parameters demonstrated a complex but physically consistent behavior across momentum, heat, and mass transfer processes. These findings provide deeper insight into the control of transport phenomena in non-Newtonian magnetohydrodynamic flows and are expected to have practical relevance in thermal engineering systems, polymer processing, and energy-related applications involving radiative heat transfer and chemically reactive flows.

Keywords: Magnetohydrodynamic flow; Jeffrey fluid; Nonlinear thermal radiation; Viscosity variation; Convective heat transfer; Boundary layer flow

1 Introduction

Non-Newtonian fluids arise naturally in numerous industrial and biological processes because the classical Newtonian fluid assumption fails to adequately capture the complex rheological behavior of many real fluids. These fluids exhibit distinctive characteristics such as shear-dependent viscosity, as well as relaxation and retardation effects. Such behaviors are commonly encountered in polymer processing, coating and extrusion operations, petroleum engineering, biomedical flows, food processing, and various geophysical applications. Consequently, the study of non-Newtonian fluid models has attracted sustained attention from researchers over the past few decades. Among the various non-Newtonian fluid models, the Jeffrey fluid model has emerged as particularly significant due to its capability to incorporate both relaxation and retardation times while preserving relative mathematical simplicity. In contrast to more complex viscoelastic models, the Jeffrey fluid provides a realistic representation of viscoelastic behaviors without involving higher-order derivatives, thereby making it especially attractive for analytical as well as numerical investigations. Moreover, the Jeffrey fluid model has been successfully employed to analyze flows over stretching surfaces, in channels, and within boundary-layer configurations pertinent to industrial manufacturing and material processing.

Recent investigations have further highlighted the significance of this model in describing viscoelastic transport phenomena arising in engineering and biomedical applications. To capture its complex rheological characteristics, researchers have explored Jeffrey fluid flow under a wide range of physical mechanisms, geometrical configurations,

and boundary conditions. These studies have incorporated additional effects such as thermal transport, mass diffusion, magnetic fields, porous media, and chemical reactions, thereby broadening the applicability of the model to realistic practical scenarios. A recent theoretical study analyzed multiphase Jeffrey fluid flow in a divergent channel and highlighted the influence of electrical double-layer effects and wall lubrication on the flow characteristics [1]. Their results underscored the critical role of electro-kinetic forces in microfluidic systems; however, thermal aspects were subsequently examined through the Cattaneo–Christov heat flux model in Jeffrey fluid flow over a stretching cylinder with internal heat generation [2]. Their findings indicated that thermal relaxation significantly modifies the temperature distribution, although nonlinear radiation and electromagnetic effects were not incorporated into their analysis. More complex transport configurations have also been explored. The peristaltic flow of a Jeffrey fluid with variable viscosity and partial slip was investigated, demonstrating the combined influence of rheological properties and wall conditions on pumping performance [3]. Flow and heat transfer in non-Newtonian fluids over an oscillating flat plate were also examined, revealing substantial alterations in the velocity and thermal boundary layers; however, that study was not specifically formulated for the Jeffrey model [4]. In recent time, the study [5] incorporated nonlinear thermal radiation into the analysis of Jeffrey nanofluid heat and mass transfer, showing that radiation significantly enhances thermal transport. Nevertheless, electromagnetic effects and coupled viscoelastic–thermal interactions were not addressed, leaving scope for further comprehensive investigation.

Viscosity is a fundamental fluid property that quantifies a fluid’s resistance to deformation under applied stress. It is commonly associated with the perceived “thickness” of a fluid and governs its response to external forces, with higher viscosity indicating greater resistance to motion. This property is primarily influenced by intermolecular forces, molecular size, and molecular structure, and it generally decreases with increasing temperature. Viscosity is very crucial in fluid motion, pressure drop, and energy dissipation in conduits and channels. Consequently, it is central to the design and optimization of fluid-based systems such as pipelines, pumps, turbines, and thermal management devices, as well as to understanding the behavior of complex fluids, including polymers, suspensions, and nanofluids. Manjunatha et al. [6] examined boundary-layer flow of hybrid nanofluids incorporating temperature-dependent viscosity and demonstrated that the viscosity ratio strongly governs velocity attenuation and thermal boundary-layer thickness. Their results indicated that a reduction in effective viscosity enhances natural convection and heat transfer rates; however, viscoelastic effects inherent in non-Newtonian models were not considered. Gbadeyan et al. [7] analyzed Casson nanofluid flow with variable viscosity and thermal conductivity, emphasizing the role of viscosity variation in modulating wall shear stress and heat transfer under convective heating and slip conditions. Although they reported that increasing viscosity variation suppresses momentum transport and significantly alters thermal gradients, the study was restricted to yield-stress fluids and excluded viscoelastic relaxation effects. Ahmed et al. [8] numerically investigated unsteady magnetohydrodynamic nanofluid flow with variable viscosity over a permeable shrinking surface, revealing a strong coupling between viscosity variation and Lorentz forces in determining boundary-layer thickness and heat transfer rates. Nevertheless, their analysis was confined to nanofluids without viscoelastic memory characteristics. Similarly, Rafique et al. [9] explored magnetohydrodynamic hybrid nanofluid flow over a stretching surface, highlighting the combined influence of viscosity ratio and velocity slip on momentum and thermal transport. The results demonstrated that increasing viscosity variation reduces flow velocity while enhancing thermal resistance near the surface; however, viscoelastic behavior and nonlinear thermal mechanisms were not addressed. Salahuddin et al. [10] incorporated variable viscosity into the flow motion of the fluid model and showed that viscosity variation plays a dominant role in controlling thermal relaxation and diffusion processes. Their results revealed suppression of temperature and concentration fields at higher viscosity ratios, though electromagnetic effects and viscoelastic retardation parameters were neglected. The study [11] demonstrated that viscosity variation significantly intensifies irreversibility and energy dissipation near channel walls. However, the analysis was limited to peristaltic configurations and did not extend to boundary-layer or external flow systems.

In addition to viscosity variation, nonlinear thermal radiation has emerged as a critical factor in nuclear reactors, thermal insulation technologies, and aerospace systems. Unlike linearized radiation models, nonlinear thermal radiation more accurately represents radiative heat flux at elevated temperatures, substantially influencing temperature distributions and thermal boundary-layer thickness. When coupled with non-Newtonian rheology, these effects generate highly nonlinear systems requiring robust numerical treatment. Salawu et al. [12] investigated magnetized hybrid Prandtl–Eyring nanofluid in aircraft applications and reported that nonlinear radiation significantly affects energy efficiency and thermal distribution, although viscoelastic and porous media effects were not included. Jha and Samaila [13] analyzed thermal radiation of a stretching sheet and showed that radiation nonlinearity markedly modifies temperature profiles and boundary-layer thickness; however, magnetic and nanofluid effects were excluded. Nasir et al. [14] examined ternary hybrid nanofluid flow with a magnetic dipole and demonstrated that nonlinear radiation enhances heat transport while altering thermal gradients, though viscoelasticity was not considered.

Porous stretching flow of Maxwell nanofluid was studied, revealing that nonlinear thermal emission, together with Dufour and Soret effects, strongly influences thermal and concentration fields; nevertheless, magnetic field effects were omitted [15]. Heat and mass transfer in Jeffrey nanofluid flow with nonlinear thermal radiation was

analyzed, demonstrating significant enhancement in temperature distribution and Nusselt number, though complex boundary-layer variations were not fully explored [16]. A Lie group approach was employed to examine electrically conducting Jeffrey nanofluid flow over a stretching sheet with nonlinear radiation, showing notable alterations in thermal boundary-layer thickness and energy dissipation rates; however, porous and additional electromagnetic effects were not incorporated [17]. Recent investigations have also emphasized the role of magnetohydrodynamic effects in Jeffrey fluid transport. In electrically conducting fluids, magnetohydrodynamics generates forces which regulate flow behavior in the system, suppress instabilities, and control heat transfer rates—features essential in metallurgical processing, polymer extrusion, cooling technologies, and energy systems. Magnetohydrodynamic electro-osmotic flow of Jeffrey fluid in a convergent geometry was modeled, highlighting the pronounced influence of magnetic forces on velocity distribution [18]. Magnetohydrodynamic free convection of Jeffrey fluid using a fractional Prabhakar framework was examined, showing that magnetic field strength significantly modifies thermal transport characteristics [19]. Cilia-driven magnetohydrodynamic Jeffrey fluid flow through porous media was investigated, demonstrating strong magnetic influence on flow distribution and pressure gradients [20]. Mathematical models of Jeffrey fluid magnetohydrodynamics over curved surfaces were conducted, reporting substantial alterations in non-similar velocity and thermal boundary layers [21]. Hall effects in magnetohydrodynamic Jeffrey fluid flow with Cattaneo–Christov heat flux were analyzed, demonstrating that electromagnetic forces critically influence both momentum and energy transport [22].

Despite these substantial contributions, most existing studies have examined variable viscosity, nonlinear thermal radiation, viscoelasticity, and magnetohydrodynamic effects separately or under simplified geometrical configurations. Comprehensive analyses integrating temperature-dependent viscosity, nonlinear radiation, viscoelastic Jeffrey rheology, and magnetic field effects within confined geometries remain scarce. In particular, the combined influence of these mechanisms in vertical channel flows—configurations of significant relevance to industrial processing and thermal management systems—has not been thoroughly investigated. To bridge this gap, the current work examines the coupled effects of Jeffrey fluid viscoelasticity, magnetohydrodynamics, nonlinear thermal radiation, and viscosity variation on temperature flow within a perpendicular channel. The proposed model offers a more accurate and practically relevant context for advanced engineering, manufacturing, and energy applications.

2 Mathematical Formulation of the Problem

This study investigates the steady-state flow of a viscous, incompressible Jeffrey fluid over a continuously stretching sheet (Figure 1), with coupled heat and mass transfer in the presence of a transverse magnetohydrodynamic field. The flow is assumed to occur along the vertical x -direction. The Jeffrey fluid model is employed to capture non-Newtonian viscoelastic behavior. To simplify the governing equations, the boundary layer approximation is adopted. A uniform transverse magnetic field is applied, and the induced magnetic field is neglected due to the low magnetic Reynolds number. Thermal radiation is modeled using the Rosseland diffusion approximation. A uniform internal heat source or sink is considered, and Joule heating is included in the energy equation, while viscous dissipation effects are neglected. Both linear and nonlinear buoyancy forces are incorporated, and mass transfer is assumed to follow Fick's law.

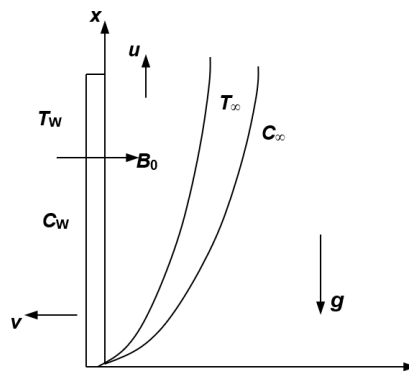


Figure 1. Stretching sheet flow

Note: u and v represent the velocity components along x and y -direction, B_0 is the magnetic field strength, T_w and T_∞ are at wall and ambient temperature, C_w and C_∞ are at wall and ambient concentrations and g is the acceleration due to gravity.

Following the formulation of the study [22] and under the assumptions, the governing equations for continuity, momentum, thermal energy, and concentration fields are presented below.

$$\frac{\partial v}{\partial y} + \frac{\partial u}{\partial x} = 0 \quad (1)$$

where, u and v are the velocity components in the x - and y -directions, respectively.

$$u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial y} = \frac{\mu_1}{\mu_f} \frac{v}{\lambda} \frac{\partial^2 u}{\partial y^2} + g\rho\beta_{C0}(C - C_\infty) + g\rho\beta_{C1}(C - C_\infty)^2 + g\rho\beta_{T0}(T - T_\infty) + g\rho\beta_{T1}(T - T_\infty)^2 - \frac{\sigma B_0^2}{\rho} u \quad (2)$$

where, μ_1 , μ_f , and ν denote the local, reference, and kinematic viscosities, respectively; λ is the Jeffrey parameter; g is the acceleration due to gravity; ρ is the density; β_{C0} and β_{C1} are the linear and nonlinear solutal expansion coefficients, respectively; β_{T0} and β_{T1} are the linear and nonlinear thermal expansion coefficients, respectively; σ is the electrical conductivity; B_0 is the magnetic field strength; T is the temperature; C is the concentration; and α is the angle of inclination.

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \frac{Q_0}{\rho c_p} (T - T_\infty) + \alpha \left[\frac{\partial^2 T}{\partial y^2} - \frac{1}{\alpha_m} \frac{\partial q_r}{\partial y} \right] + \frac{\sigma B_0^2}{\rho} u^2 \quad (3)$$

where, Q_0 is the non-uniform heat generation parameter; c_p is the specific heat; q_r is the radiative heat flux.

$$u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} = D \frac{\partial^2 C}{\partial y^2} - K_0 (C - C_\infty) \quad (4)$$

where, C_∞ is the ambient concentration and K_0 is the chemical reaction parameter.

Subject to boundary conditions stated as follow:

$$\begin{aligned} v(0) &= 0, u(0) = U_\infty \\ -k \frac{\partial T}{\partial y}(0) &= h(T_\infty - T(0)) \\ T(0) &= 0, T(\infty) = T_\infty \\ C(\infty) &= C_\infty, C_w(0) = Ax^\omega + C_\infty \end{aligned} \quad (5)$$

where, h is the convective heat transfer coefficient; k is the thermal conductivity.

The stream function is presented as follow:

$$\begin{aligned} v &= -\frac{\partial \psi}{\partial x} \\ u &= \frac{\partial \psi}{\partial y} \end{aligned} \quad (6)$$

The following dimensionless parameters are used.

$$\begin{aligned} Q &= \frac{Q_0}{\rho c_p}, \text{Pr} = \frac{\nu}{\alpha}, M = \frac{\sigma B_0^2}{\rho}, \text{Bi} = \frac{h}{k} \left(\frac{vx}{u_0} \right)^{\frac{1}{2}}, \phi = \frac{C - C_\infty}{C_w - C_\infty}, \\ \gamma &= \frac{k_1 x}{u_0}, \theta = \frac{T - T_\infty}{T_w - T_\infty}, y = \sqrt{\frac{C_1}{\nu}} \frac{y}{v}, \mu = \frac{\mu_1}{\mu_f}, \delta_2 = \frac{\beta_{C1}(C_w - C_\infty)}{\beta_{C0}}, \\ x &= \frac{C_1}{U_1}, u = \frac{u}{U_1}, \text{Gc} = \frac{g\beta(C_w - C_\infty)x}{u_\infty^2}, R = \frac{4\sigma(T_w - T_\infty)}{k^* k_1}, \\ \text{Sc} &= \frac{\nu}{D}, \delta_1 = \frac{\beta_{T1}(C_w - C_\infty)}{\beta_{T0}}, C_t = \frac{T_\infty}{T_w - T_\infty}, \text{Gr} = \frac{g\beta T(T_w - T_\infty)}{u_\infty^2} \end{aligned} \quad (7)$$

where, Q is the heat source/sink parameter; Pr is the Prandtl number; M is the magnetic parameter; Bi is the Biot number; ϕ is the concentration parameter; γ is the chemical reaction parameter; μ is the viscosity variation parameter; δ_2 is the density variation parameter with concentration; Gc is the local solutal Grashof number; R is the nonlinear thermal radiation parameter; Sc is the Schmidt number; δ_1 is the density variation parameter with temperature; C_t is the temperature variation parameter; and Gr is the local thermal Grashof number.

Substituting Eqs. (6)–(7) into Eqs. (1)–(5) leads to the following equations:

$$\begin{aligned} (1 + \lambda) \left[\frac{\partial \psi}{\partial y} \frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial \psi}{\partial x} \frac{\partial^2 \psi}{\partial y \partial x} \right] &= \mu \frac{\partial \psi}{\partial x} \frac{\partial^3 \psi}{\partial y^2 \partial x} - (1 + \lambda) \text{Mu} \\ &+ (1 + \lambda) \text{Gr} [\theta + \delta_1 \theta^2] + (1 + \lambda) \text{Gc} [\phi + \delta_2 \phi] \end{aligned} \quad (8)$$

$$\begin{aligned} \text{Pr} \left[\frac{\partial \psi}{\partial y} \frac{\partial \theta}{\partial x} - \frac{\partial \psi}{\partial x} \frac{\partial \theta}{\partial y} \right] &= \left[1 + \frac{4R}{3} (C_t + \theta)^3 \right] \frac{\partial^2 \theta}{\partial y^2} \\ &+ 4 [C_t + \theta]^2 \left(\frac{\partial \theta}{\partial y^2} \right)^2 + Q\theta + \text{Mu}^2 \end{aligned} \quad (9)$$

$$\frac{\partial \psi}{\partial y} \frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial x} \frac{\partial \phi}{\partial y} = \frac{\partial^2 \phi}{\partial y^2} - \gamma \text{Sc} \phi \quad (10)$$

The boundary conditions are given below:

$$\begin{aligned} \frac{\partial \psi}{\partial y} = x, -\frac{\partial \psi}{\partial x} = \frac{V_w}{\sqrt{C_1 v}}, \frac{\partial \theta}{\partial y} = -\text{Bi}[\theta(0) - 1], \phi = 1, \\ \theta(0) = 0, \phi(0) = 0, \frac{\partial \psi}{\partial y}(\infty) = 0 \end{aligned} \quad (11)$$

The presentation below shows the similarity variables:

$$\begin{aligned} \eta = y \left(\frac{u_0}{v_x} \right)^{\frac{1}{2}}, \psi = \sqrt{xvU_0 f(\eta)} \\ \theta(\eta) = \frac{T_f - T_\infty}{T_i - T_\infty}, \phi(\eta) = \frac{C - C_\infty}{C_w - C_\infty} \end{aligned} \quad (12)$$

Substituting Eq. (12) into Eqs. (8)–(11) leads to the following equations:

$$\mu A''' + \frac{1}{2}(1 + \lambda)AA'' - (1 + \lambda)MA' + (1 + \lambda) \text{Gr} [\theta + \delta_1 \theta^2] + (1 + \lambda) \text{Gc} [\phi + \delta_2 \phi] = 0 \quad (13)$$

$$\theta'' \left[1 + \frac{4}{3} R (\theta + C_t)^3 + 4R [C_t + \theta]^2 [\theta'] + \frac{1}{2} \text{Pr} \theta' A + Q\theta + \text{Mu}^2 = 0 \right] \quad (14)$$

$$\phi'' + \frac{1}{2} \text{Sc} A \phi' - \text{Sc} \gamma \phi = 0 \quad (15)$$

where, $\eta, A(\eta), \theta(\eta)$, and $\phi(\eta)$ are similarity constants. The problem is solved subject to the following boundary conditions:

$$\begin{aligned} A(0) = 0, A'(0) = 1, \theta'(0) = \text{Bi}[\theta(0) - 1], \phi(0) = 1, \\ A'(\infty) = 0, \theta(0) = 0, \phi(0) = 0 \end{aligned} \quad (16)$$

2.1 Physical Quantities

In order to underscore the engineering importance, several derived quantities are evaluated, namely, skin friction, Nusselt number, and Sherwood number, as follows:

$$C_f = \frac{1}{2} \text{Re}_x^{-\frac{1}{2}}, C_f = f''(0) \quad (17)$$

$$\text{Nu} = \text{Re}_x^{-\frac{1}{2}}, \text{Nu} = -\theta'(0) \quad (18)$$

$$\text{Sh} = \text{Re}_x^{-\frac{1}{2}}, \text{Sh} = -\phi'(0) \quad (19)$$

2.2 Numerical Techniques

The MATLAB bvp4c solver is a robust and efficient numerical technique for solving boundary value problems arising from systems of ordinary differential equations. In the present study, it is utilized to compute the solutions of Eqs. (13)–(15) subject to the boundary conditions specified in Eq. (16). The solver employs a collocation method based on a fourth-order Runge–Kutta scheme with adaptive mesh refinement, thereby ensuring high computational accuracy and stability. A step size of 0.01 and a convergence tolerance of 10^{-10} are adopted to obtain reliable numerical results. The computational procedure involves the following steps:

Step 1: Reformulating the governing equations into a corresponding method of first-order differential equations:

$$\begin{aligned} A = F(1), A' = F(2), A'' = F(3), A''' = F'(3), \theta = F(4), \\ \theta' = F(5), \theta'' = F'(5), \phi = F(6), \phi' = F(7), \phi'' = F'(7) \end{aligned} \quad (20)$$

Step 2: Ensuring that the number of first-order equations is consistent with that of boundary conditions.

Step 3: Expressing the boundary conditions in residual form:

$$\begin{bmatrix} \text{Fa}(1), \text{Fa}(2) - 1, \text{Fa}(5) - \text{Bi} \times (\text{Fa}(4) - 1), \text{Fa}(6) - 1, \\ \text{Fb}(2), \text{Fb}(4), \text{Fb}(6) \end{bmatrix} \quad (21)$$

Step 4: Solving the resulting system numerically using the bvp4c algorithm.

2.3 Validation

The accuracy of the present numerical formulation is validated by considering a limiting case in which the magnetic field and heat source parameters are set to zero. Under these simplified conditions, the governing equations reduce to those reported by Samaila and Jha [22]. Table 1 shows the evaluation of $-\theta'(\eta)$ and $-\phi'(\eta)$ for diverse values of Pr. The excellent agreement between the present results and the previously published data confirms the accuracy, validity, and reliability of the present numerical scheme.

Table 1. Numerical comparison of the temperature ($-\theta(\eta)$) and the concentration ($-\phi''(\eta)$) with the results reported by Samaila and Jha [22] for selected values

Pr	Samaila and Jha [22]		Present Results	
	$-\theta'(\eta)$	$-\phi'(\eta)$	$-\theta'(\eta)$	$-\phi'(\eta)$
0.044	0.041092	0.671516	0.041092	0.671516
0.720	0.076277	0.662266	0.076277	0.662266
2.000	0.086772	0.661034	0.086772	0.661034

Note: Computations were performed for $Gc = Gr = Bi = 0.1$, $M = Q = 0$, $Sc = 0.62$, $N = 2$, $\delta_2 = 0.2$, $\gamma = 0.5$, $\delta_1 = 0.3$, $C_t = 0.2$.

3 Results and Discussion

This study examines the influence of nonlinear thermal radiation on magnetohydrodynamic Jeffrey fluid flow over a stretching sheet. The governing equations consist of a system of nonlinear partial differential equations, which are transformed into a dimensionless form through the application of appropriate similarity transformations. The resulting system of ordinary differential equations is solved subject to the prescribed boundary conditions. Unless otherwise stated, the default values of the governing physical parameters are specified below.

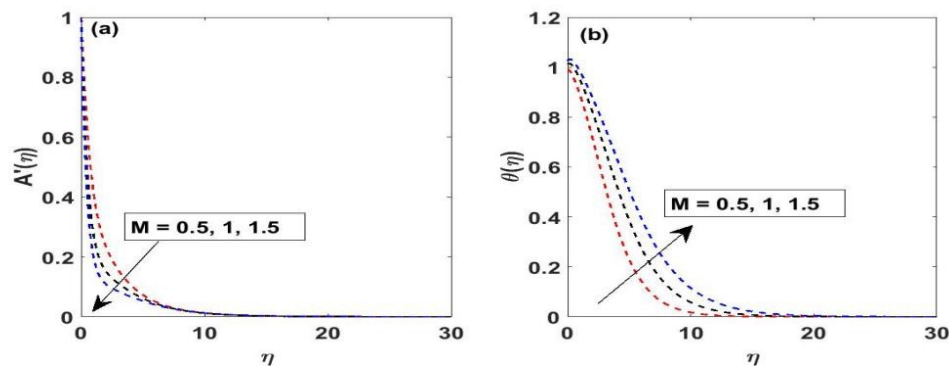


Figure 2. Influence of the magnetic field parameter (M) on: (a) the velocity $A'(\eta)$; (b) the temperature $\theta(\eta)$ profiles

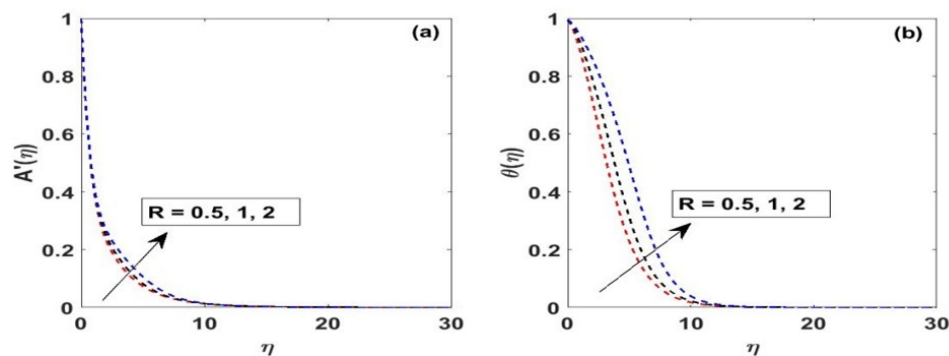


Figure 3. Influence of the nonlinear thermal radiation parameter (R) on: (a) the velocity $A'(\eta)$; (b) the temperature $\theta(\eta)$ profiles

Figure 2 and Figure 3 depict the influence of the magnetic field parameter (M) and the nonlinear thermal radiation parameter (R) on the velocity and temperature distributions of the Jeffrey fluid. As illustrated in Figure 2a, an upsurge in M results in a pronounced decline in the outline throughout the boundary layer. When a conducting fluid moves

in the presence of a magnetic field, the motion of charged particles (ions and electrons) generates an electric current. This current interacts with the magnetic field and produces a resistive force known as the Lorentz force. Physically, this force acts opposite to the direction of fluid motion. Figure 2b displays that the temperature profile upsurges with enhancing values of M , accompanied by a corresponding thickening of the thermal boundary layer. This trend is due to the magnetic field present in the heat equation, which produces additional internal heat within the fluid. The conversion of electrical energy into thermal energy elevates the fluid temperature and diminishes the rate of heat dissipation from the surface.

The influence of R on the velocity distribution is presented in Figure 3a. An increase in R slightly enhances the velocity profile and marginally thickens the momentum boundary layer. This behavior can be credited to the rise in fluid temperature resulting from enhanced radiative heat transfer, which effectively reduces fluid viscosity and weakens viscous resistance to flow. Figure 3b further demonstrates that the temperature profile increases significantly with increasing R , leading to a substantial growth in thermal boundary layer thickness. R modifies the thermal boundary layer by significantly increasing the effective thermal conductivity of the fluid. As the temperature rises, radiative heat transfer becomes stronger. This results in faster heat diffusion away from the surface and higher fluid temperatures within the boundary layer.

Figure 4a illustrates that an upsurge in the viscosity variation parameter (μ) leads to a pronounced reduction in the velocity profile, accompanied by a thicker of the momentum boundary layer due to enhanced viscous resistance. This behavior is physically attributed to the enhancement of internal resistance inside the fluid as viscosity increases. When viscosity is higher, the fluid offers greater resistance to deformation. This increases the rate of momentum diffusion (transfer of momentum between fluid layers), leading to stronger damping of fluid motion, lower velocity profiles, and a thicker momentum boundary layer due to enhanced viscous resistance. In Figure 4b, increasing μ also results in a reduction in the temperature profile and a corresponding decline in the thickness of the thermal boundary layer. The attenuation of the thermal field can be linked to weakened convective heat transport caused by the reduced fluid velocity. Since convective mechanisms become less effective under stronger viscous effects, the overall heat distribution within the boundary layer diminishes.

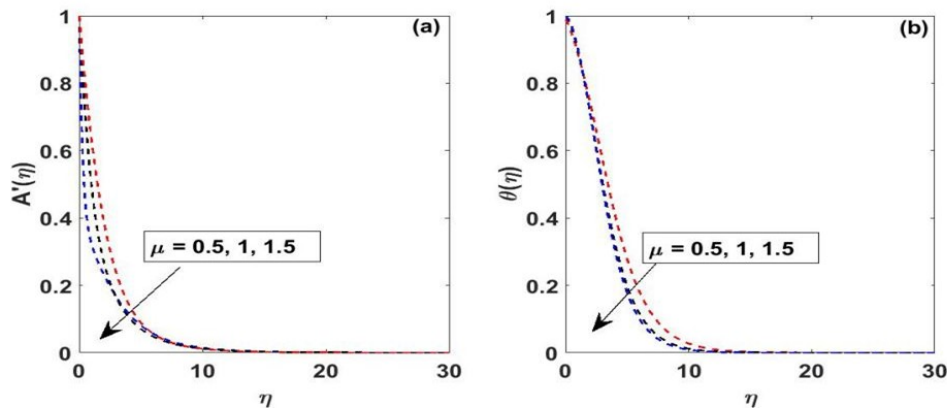


Figure 4. Influence of the viscosity variation parameter (μ) on: (a) the velocity $A'(\eta)$; (b) the temperature $\theta(\eta)$ profiles

The quantitative effects of the magnetic field parameter (M), heat source parameter (Q), Jeffrey parameter (λ), viscosity ratio (μ), and nonlinear thermal radiation parameter (R) on the engineering quantities of interest are summarized in Table 2. An increase in M , λ , μ , and Q leads to a decline in the skin-friction coefficient, signifying a decrease in wall shear stress. In contrast, the nonlinear thermal radiation parameter enhances the skin friction coefficient. This trend reflects the competing influences of electromagnetic damping, viscoelastic resistance, viscous effects, and radiative heating on surface drag characteristics. With respect to heat transfer, the magnetic field parameter reduces the Nusselt number, signifying a decline in the surface heat transfer rate due to magnetic damping and Joule heating effects. Conversely, the heat source parameter, Jeffrey parameter, viscosity ratio, and nonlinear thermal radiation parameter enhance the Nusselt number, demonstrating that internal heat generation, viscoelastic properties, viscous variation, and radiative energy transport significantly influence the thermal performance of the system. Similarly, the Sherwood number is suppressed by increasing values of the magnetic field parameter, heat source parameter, Jeffrey parameter, and viscosity ratio, indicating a reduction in surface mass transfer rate. However, the nonlinear thermal radiation parameter exhibits an opposite effect by enhancing the Sherwood number. These findings collectively demonstrate that electromagnetic forces, internal heat generation, viscoelastic rheology, viscosity variation, and nonlinear radiative effects play pivotal and interdependent roles in regulating surface shear

stress, heat transfer, and mass transport in the present magnetohydrodynamic Jeffrey fluid configuration.

Table 2. Effects of magnetic field parameter (M), heat source/sink (Q), viscosity variation (μ), and nonlinear thermal radiation parameter (R) on skin friction coefficient (C_f), Nusselt number (Nu), and Sherwood number (Sh)

M	Q	λ	μ	R	C_f	Nu	Sh
0.5	0.01	0.5	0.5	0.5	-0.63218	0.00964	0.24927
1.0					-1.03233	-0.08404	0.24364
2.0					-1.58348	-0.18879	0.23849
0.5	0.01				-0.63218	0.00964	0.24927
	0.10				-0.68525	0.13834	0.24491
	0.20				-0.70415	0.17720	0.24385
	0.01	0.5			-0.63218	0.00964	0.24927
		0.8			-0.68456	0.01333	0.24856
		1.0			-0.71696	0.01535	0.24817
		0.5	0.5		-0.63218	0.00964	0.24926
			0.8		-0.68456	0.01333	0.24856
			1.0		-0.70281	0.01534	0.24817
			0.5	0.5	-0.63218	0.00963	0.24927
				1.0	-0.62536	0.01339	0.24978
				2.0	-0.61706	0.01454	0.25055

Figure 5a and Figure 5b depict the influence of Gr on the velocity and temperature distributions, respectively. Increasing Gr significantly enhances the velocity in the vicinity of the surface. Since Gr quantifies the proportion of buoyancy forces to viscous forces, larger values of Gr indicate stronger buoyancy-driven effects. The intensified buoyancy force accelerates the velocity of the fluid, thereby enhancing the momentum boundary-layer thickness and promoting stronger natural convection within the flow field. Similarly, the temperature profile decreases with enhancing Gr , as illustrated in Figure 5b. The enhancement of buoyancy-induced flow strengthens convective heat transfer away from the surface, leading to more efficient thermal energy removal. Consequently, the thermal boundary layer becomes thinner, and the fluid temperature decays more rapidly with the similarity variable η . This behavior underscores the dominant role of natural convection in regulating both momentum and heat transfer characteristics at higher Grashof numbers, highlighting the strong coupling between buoyancy forces and thermal transport in the present magnetohydrodynamic Jeffrey fluid system.

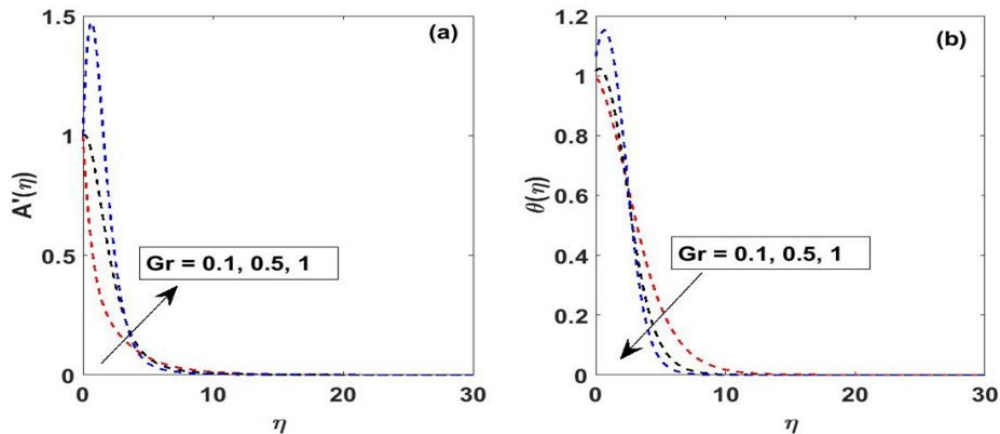


Figure 5. Influence of local thermal Grashof (Gr) on: (a) the profile velocity $A'(\eta)$; (b) the temperature $\theta(\eta)$ profiles

Figure 6a and Figure 6b illustrate the effects of the Jeffrey parameter (λ) on the velocity and temperature distributions. Therefore, an increase in λ reduces both temperature and velocity profiles all over the boundary layer. Actually, increasing the Jeffrey parameter emphasizes fluid relaxation effects relative to retardation effects, thereby increasing the effective viscoelastic resistance to flow. This additional friction suppresses the fluid motion, leading to a thinner momentum boundary layer. Similarly, the temperature profile decreases with upsurging λ , reflecting diminished convective heat transport and declined thermal boundary-layer thickness. Consequently, larger values of

λ correspond to lower fluid temperatures within the domain. The impact of the temperature difference parameter C_t on velocity and temperature fields is depicted in Figure 7a and Figure 7b. Increasing C_t slightly suppresses the velocity profile, as larger thermal gradients enhance fluid density variations and modify the balance between buoyancy and viscous forces, resulting in a marginal reduction in flow speed. In addition, the temperature profile upsurges significantly with increasing C_t , indicating stronger thermal driving at the wall. This elevates the heat of the fluid and the thermal boundary layer stiffness, promoting improved diffusion of thermal energy into the fluid.

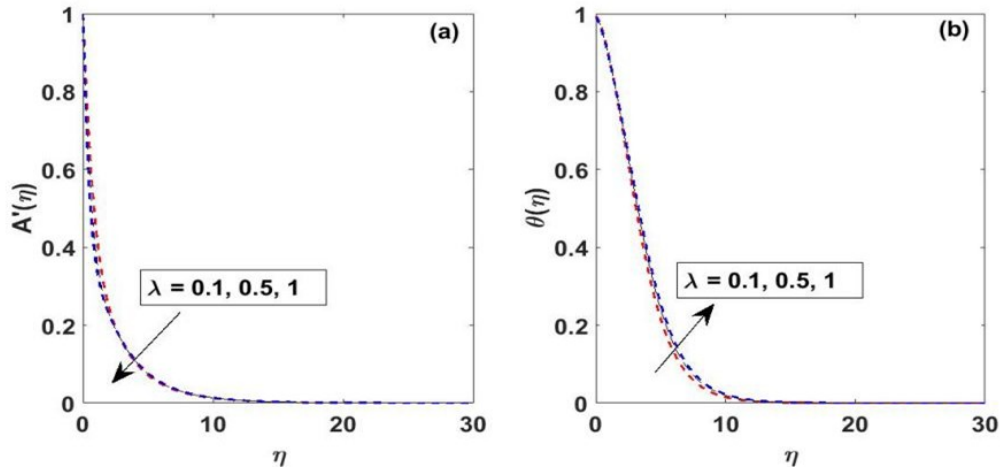


Figure 6. Influence of the Jeffrey parameter (λ) on: (a) the velocity $A'(\eta)$; (b) the temperature $\theta(\eta)$ profiles

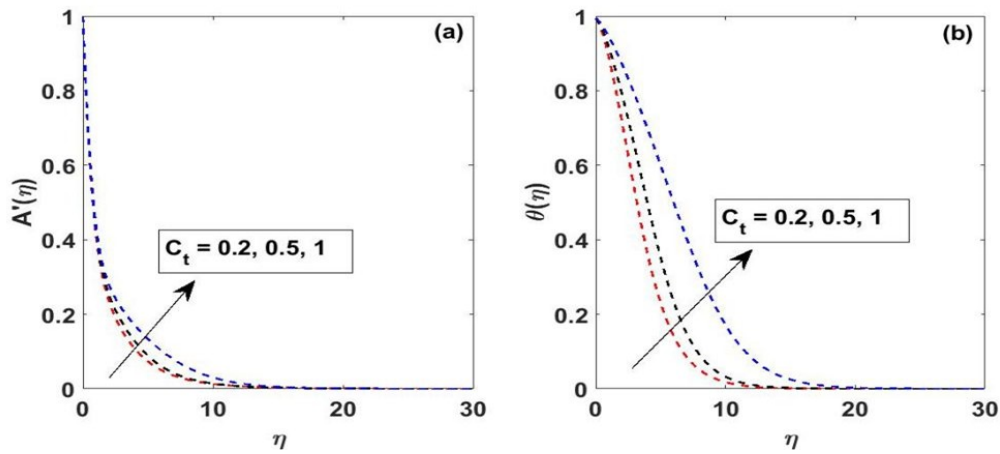


Figure 7. Influence of the Jeffrey parameter (C_t) on: (a) the velocity $A'(\eta)$; (b) the temperature $\theta(\eta)$ profiles

Figure 8a and Figure 8b depict the upshots of the Schmidt number (Sc) on velocity and concentration outcomes. As shown in Figure 8a, an increase in Sc slightly reduces the velocity within the boundary layer, as lower mass diffusivity weakens buoyancy-driven acceleration arising from concentration gradients, leading to a marginally thinner momentum boundary layer. Figure 8b demonstrates that higher Sc values strongly suppress the concentration distribution. Physically, a larger Sc corresponds to reduced molecular diffusivity, restricting species transport and producing a thinner solutal boundary layer, with faster decay of concentration away from the wall. The effects of the chemical reaction parameter γ on velocity and concentration fields are presented in Figure 9a and Figure 9b. Increasing γ reduces the fluid velocity, as stronger chemical reactions diminish species concentration, weakening the solutal buoyancy contribution to flow and resulting in a thinner momentum boundary layer. Additionally, higher γ significantly decreases the concentration profile by accelerating the consumption of diffusing species, thereby reducing the solutal boundary-layer thickness. These results highlight the dominant role of chemical reactions in controlling mass transport in reactive magnetohydrodynamic Jeffrey fluid flows.

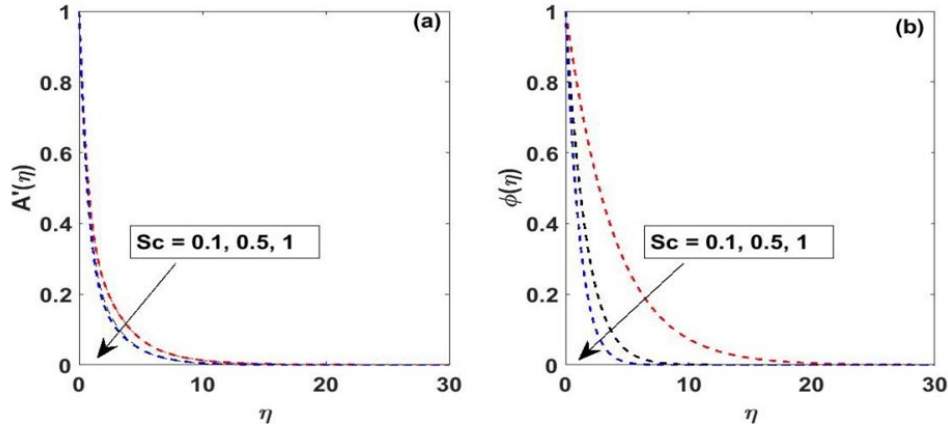


Figure 8. Influence of the Jeffrey parameter (Sc) on: (a) the velocity $A'(\eta)$; (b) the temperature $\phi(\eta)$ profiles

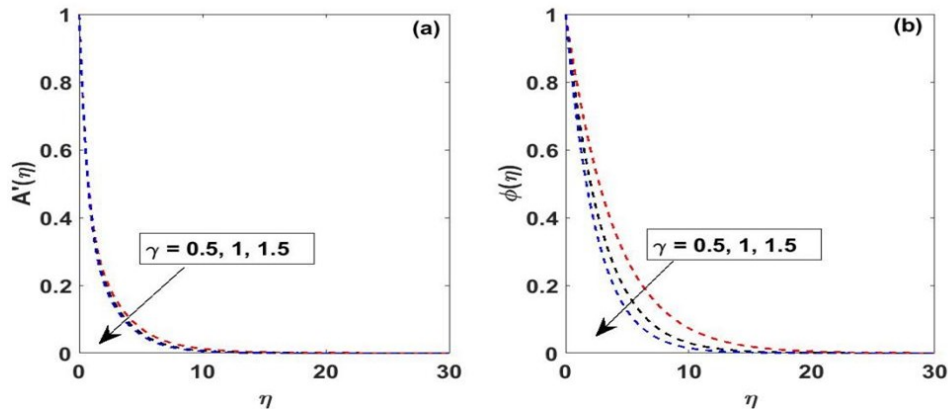


Figure 9. Influence of the Jeffrey parameter (γ) on: (a) the velocity $A'(\eta)$; (b) the temperature $\phi(\eta)$ profiles

4 Conclusions

This study examined steady heat and mass transfer in an electrically conducting magnetohydrodynamic Jeffrey fluid over a stretching sheet, incorporating viscosity variation effects, nonlinear thermal radiation, internal heat generation, and chemical reaction. The governing equations were transformed using similarity variables and solved numerically to analyze velocity, temperature, and concentration fields. The results indicate that:

- The magnetic field suppresses fluid velocity via the Lorentz force while enhancing temperature through Joule heating.
- Nonlinear thermal radiation increases both velocity and temperature profiles, thickening the thermal boundary layer and elevating wall shear stress.
- Higher viscosity variation and Jeffrey parameters reduce velocity and temperature due to enhanced viscous and viscoelastic resistance.
- Higher Grashof numbers accelerate the flow and thin the thermal boundary layer via buoyancy-driven convection.
- Chemical reactions strongly suppress concentration and velocity, demonstrating their dominant effect on mass transport.

The present results are relevant to engineering applications involving electrically conducting non-Newtonian fluids, such as polymer processing, cooling of electronic devices, and metallurgical operations. They also provide useful insight into optimizing thermal transport and flow control in systems where magnetic fields and radiative effects are significant.

Author Contributions

Conceptualization, A.B.D.; methodology, W.M.; software, W.M. and S.O.A.; formal analysis, A.B.D.; investigation, C.Y.I., A.B.D., and S.O.A.; writing—original draft preparation, A.B.D., W.M., C.Y.I., and S.O.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Acknowledgements

The authors gratefully acknowledge the financial support provided by the Senate Research Grant of the National Open University of Nigeria for carrying out this research.

Conflicts of Interest

The authors declare no conflicts of interest.

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